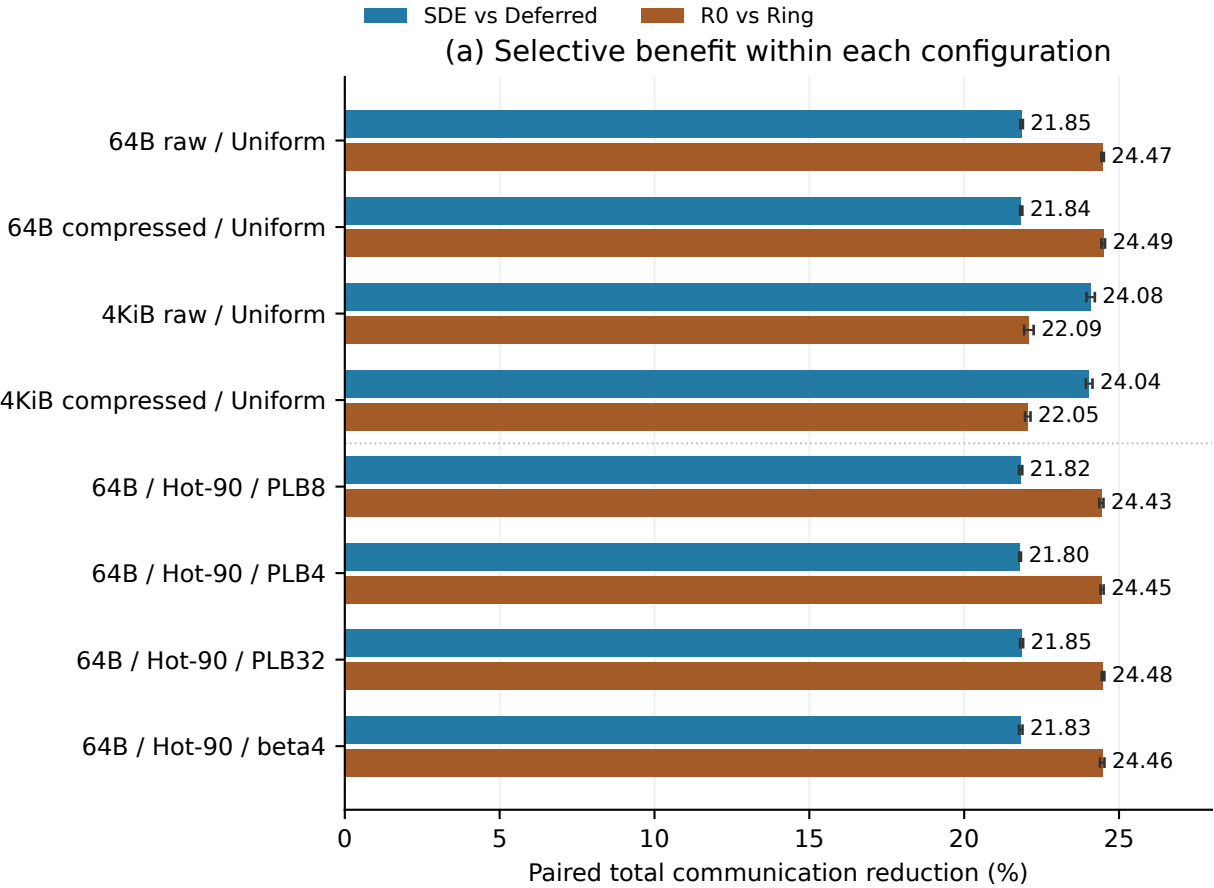
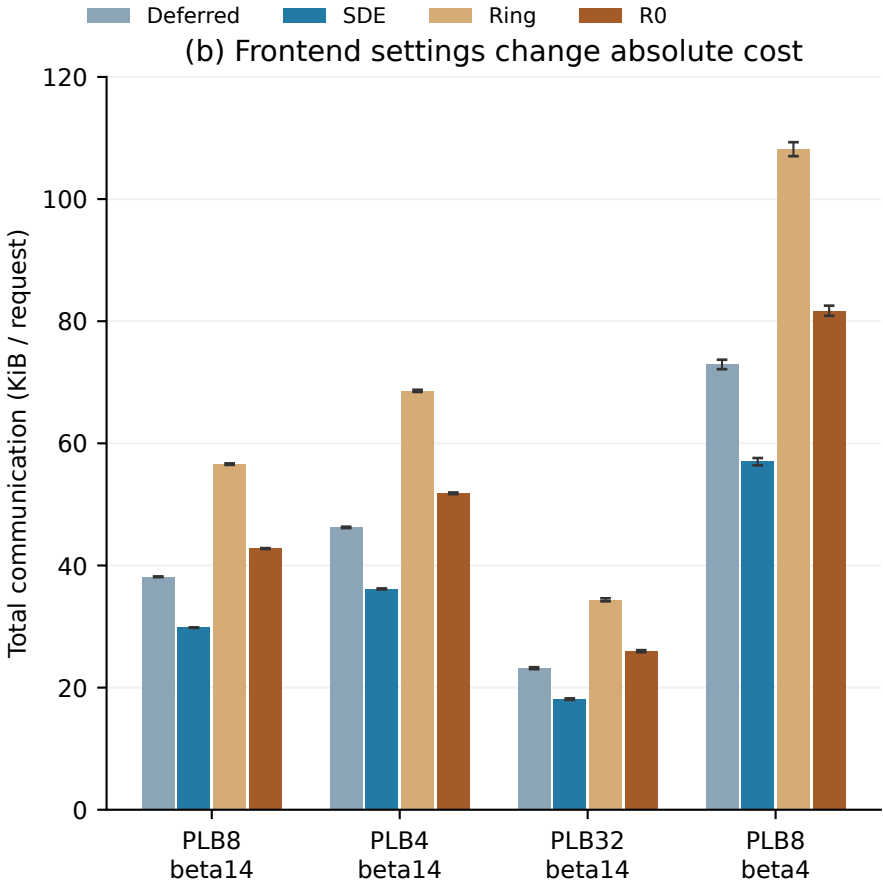


Freerecursive-style composition: selective gains and frontend costs



N = 4,096; Z4/A3/S4; R = 256; 2,048 warmup + 4,096 measured requests; five paired seeds, mean and 95% t intervals. All serialized protocol bytes, including authentication and maintenance. Panel (b): 64B compressed map, Hot-90, X = 32.



At 4KiB, X = 1,024 (raw) or 2,048 (compressed); at 64B, X = 16 or 32. Block-size slices also change packing and PLB bytes. PLB changes use different client resources. Finite windows; no native PMMAC/hardware controller, peak-memory or latency claim.